CS408

Introduction to Human-Computer Interaction

HCI is the study of how humans interact with computers and other digital systems, focusing on designing user-friendly interfaces that enhance usability and user experience.

Learning Goals

- Understanding fundamental principles of HCI.
- Exploring the interaction between humans and computers.
- Recognizing challenges and advancements in the field.

Riddles for the Information Age

This concept refers to the challenges and paradoxes posed by modern technology, such as balancing usability with security, or automation with human control.

Role of HCI

HCI addresses these riddles by designing systems that align with human needs, ensuring technology remains accessible and efficient.

Definition of HCI

HCI is a multidisciplinary field concerned with the design, evaluation, and implementation of interactive computing systems for human use.

Reasons for Non-Bright Aspects

Despite its benefits, HCI faces challenges such as:

- **Over-reliance on technology** leading to reduced human skills.
- **Privacy concerns** due to data misuse.

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- **Inequality in accessibility**, such as software apartheid (discussed below).

Human Versus Computer

This explores the contrast between human abilities (e.g., creativity, emotional intelligence) and computer capabilities (e.g., speed, precision), emphasizing the need for complementary interaction.

Software Apartheid

A term highlighting unequal access to technology, where certain groups are excluded due to economic, social, or geographic barriers.

An Industry in Denial

This reflects the reluctance of tech industries to acknowledge issues like user privacy violations, unethical AI use, or unsustainable practices.

Techno-Rage

Refers to frustration caused by poorly designed systems or technological failures, emphasizing the importance of user-centric design.

Success Criteria in the New Economy

In a digital economy, success is driven by:

- Usability and accessibility of technology.
- Innovation in interactive systems.
- Responsiveness to user feedback.

Computer + Information

This explores the synergy between computing power and information management, forming the backbone of HCI.

Goals & Evolution of Human-Computer Interaction

Goals of HCI

- Enhancing usability and user satisfaction.
- Designing systems that accommodate human limitations and strengths.
- Facilitating seamless human-computer interaction.

Evolution of HCI

- Early focus on hardware interaction.
- Shift to graphical user interfaces (GUIs).
- Emergence of touch, voice, and gesture-based interactions.

Discipline of Human-Computer Interaction

Learning Goals

- Explore the principles of interface design.
- Understand the interdisciplinary nature of HCI.

Quality

Quality in HCI is defined by factors like usability, reliability, and efficiency of interaction.

Interdisciplinary Nature of HCI

HCI draws from:

- Psychology (understanding user behavior).
- Computer Science (developing systems).
- Design (creating intuitive interfaces).

Cognitive Frameworks

Learning Goals

- Understanding how humans process information.

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- Applying cognitive models to interface design.

Introduction

Cognitive frameworks in HCI focus on human thinking patterns and how they influence interaction with technology.

Modes of Cognition

- **Automatic cognition**: Routine tasks requiring little thought.
- **Controlled cognition**: Tasks requiring active problem-solving.

Human Processor Model

A model that views humans as processors of information with limits on memory, attention, and reaction time.

GOMS

A predictive model (Goals, Operators, Methods, Selection rules) used to analyze user interaction and efficiency.

Recent Developments in Cognitive Psychology

Advances like understanding multitasking, cognitive load, and decision-making inform HCI designs.

External Cognition

How people use external tools (like notes or diagrams) to extend cognitive capabilities.

Distributed Cognition

Examines how cognitive processes are shared across individuals, artifacts, and environments.

Input-Output Channels

Vision

Vision is a primary input channel in HCI, and interfaces must consider factors like color perception, readability, and visual attention.

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Visual Perception

Visual perception is how humans interpret visual stimuli, crucial for designing user interfaces.

Learning Goals

- Understand how users perceive and process visual elements.
- Apply principles of visual design to improve interface usability.

Key Concepts

1. **Color Theory**
    - Explains how colors interact and affect user perception.
    - **Primary Colors**: Red, blue, yellow.
    - **Complementary Colors**: Colors opposite each other on the color wheel.
    - **Design Principles**: Use contrast for readability and harmony for aesthetics.
2. **Stereopsis**
    - Refers to depth perception from binocular vision.
    - Applications include 3D interfaces and augmented reality.
3. **Reading**
    - Human reading behavior influences font choice, spacing, and layout.
    - Key principle: Optimize for readability with appropriate fonts and text size.

Hearing

- Involves perceiving auditory cues.
- Applications: Notifications, voice interfaces, and auditory feedback.

Touch

- Perception of tactile feedback, important in touchscreens and haptic devices.

Movement

- Interaction based on gestures or physical movements, used in systems like Kinect or VR.

Cognitive Process

Learning Goals

- Understand how users think and process information.
- Apply cognitive psychology to design user-friendly interfaces.

Key Topics

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1. **Attention**
    - Humans have limited attention; design should focus on minimizing distractions.
2. **Memory**
    - **Revised Memory Model**: Considers short-term, long-term, and working memory.
    - Design Implications: Reduce memory load by providing cues and prompts.

Learning

Key Skills

1. **Reading, Speaking, and Listening**
    - Interfaces should accommodate diverse user preferences for consuming information.
2. **Problem Solving, Planning, Reasoning, and Decision-Making**
    - Design tools to support complex tasks, e.g., dashboards for data analysis.

The Psychology of Actions

- Explores how users translate intention into action and the role of feedback.

Mental Model

- The user's understanding of how a system works, which guides interaction.
- Design should align with users' mental models for intuitive use.

Errors

- Errors occur when the system doesn't align with user expectations.
- **Design Principle**: Provide clear error messages and recovery options.

Design Principles

Learning Goals

- Recognize and apply foundational design principles.
- Create effective, user-friendly interfaces.

The Computer

Learning Goals

- Explore the hardware and software that enable HCI.

Input Devices

- **Text Entry Devices**: Keyboards, on-screen keyboards.
- **Positioning, Pointing, and Drawing**: Mice, styluses, touchpads.

Display Devices

- Monitors, VR headsets, augmented reality displays.

Touch, Feel, and Smell

- Innovations include tactile feedback, smell-based interfaces for immersive experiences.

Physical Controls

- Buttons, sliders, knobs in physical devices.

Environment and Bio-Sensing

- Integration of sensors for temperature, motion, and physiological data.

Interaction

Learning Goals

- Understand the dynamics of human-computer interaction.

Key Concepts

1. **The Terms of Interaction**: Inputs, outputs, feedback.
2. **Donald Norman's Model**: Explains user actions through goals, intentions, and execution.
3. **The Interaction Framework**: Breaks interaction into communication between user and system.

Interaction Styles

- Command-line interfaces, menus, WIMP (Windows, Icons, Menus, Pointers), natural user interfaces (NUIs).

Interaction Paradigms

- Innovations like touch, voice, and augmented reality.

HCI Process and Models

Learning Goals

- Understand lifecycle models and methodologies in HCI design.

Lifecycle Models in HCI

1. **Goal-Directed Design**
    - A user-centered approach focusing on achieving user goals.
    - **Process Overview**: Research → Design → Prototype → Evaluate.

Types of Users

- Primary, secondary, tertiary users with varying needs and interaction levels.

User Research

Learning Goals

- Learn research methods to understand user behavior.

Qualitative Research

- Methods include interviews, focus groups, and observation.

User-Centered Approach

- Focuses on designing systems around user needs and preferences.

Ethnography Framework

- In-depth studies of users in their natural environments.

Preparing for Ethnographic Interviews

- Develop clear objectives and conduct pre-interview research.

Putting a Plan Together

- Align research goals with project requirements and user needs.

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